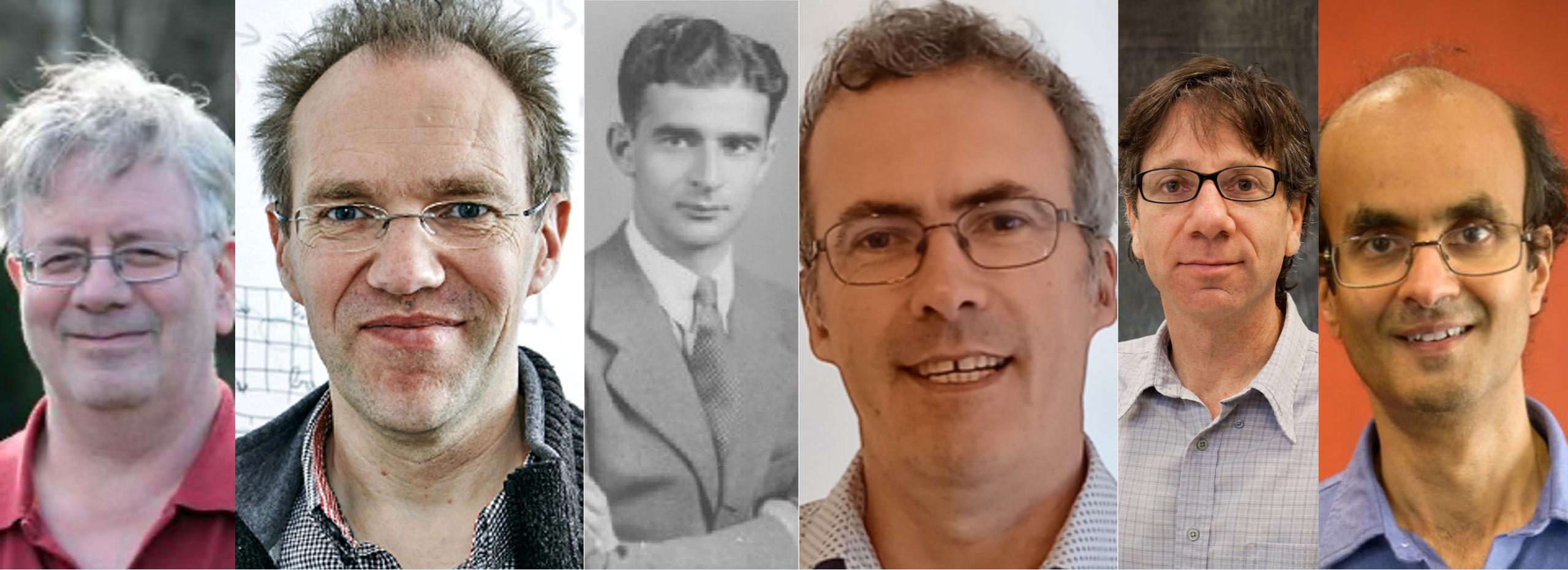


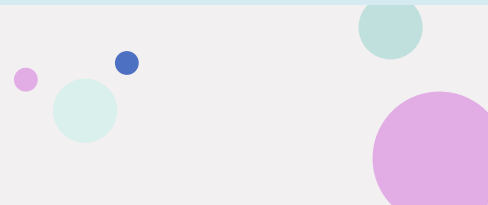


Lecture 7

CPA security, Modes of Operation

CS409: Introduction to Cryptography

Instructor: Sruthi Sekar



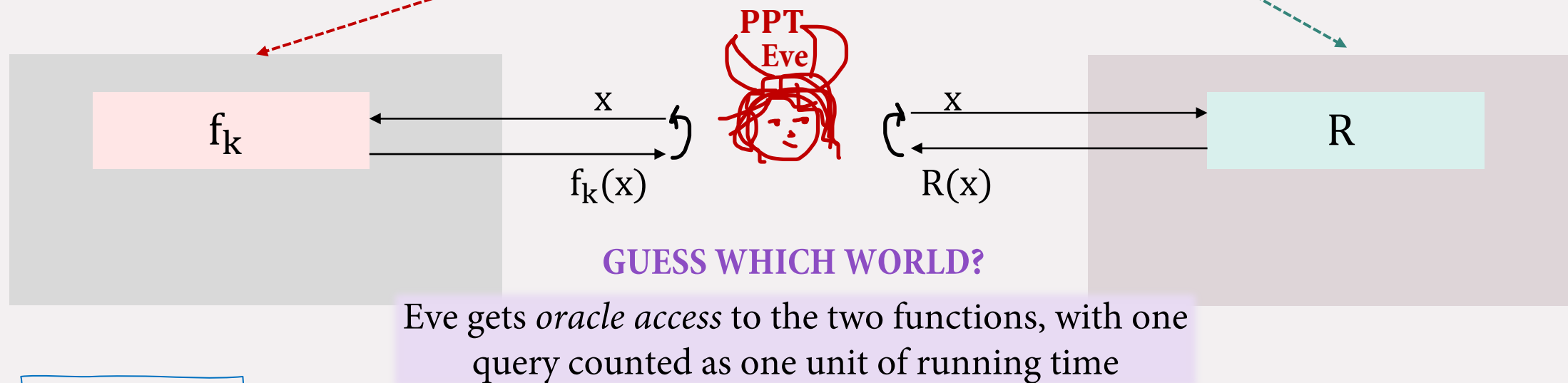
Recap

- How to encrypt multiple messages using the same key in the eavesdropper Eve world?
 - **Stateful Setting:** can be done using PRGs (not desired)
 - **Stateless:** Repeating a deterministic encryption using the same key for multiple messages is not secure.
- **Stateless Symmetric Encryption Scheme for multiple messages:** using a random function oracle (gives you exponential one time pads that are efficiently accessible)
- **Pseudorandom Functions:** computational analogue of a random oracle.

Recall: Pseudorandom Functions



Want to define a function (taking a secret key) that “seems like” a random function oracle to **PPT** Eve.



Definition 1

A family of functions $\{f_k: \{0,1\}^n \rightarrow \{0,1\}^n\}_{k \in \{0,1\}^n}$ is a **pseudorandom function** if for every **PPT Eve**, there exists a negligible function **negl** such that, $\forall n$

$$\left| \Pr_{k \in_R \{0,1\}^n} [\text{Eve}^{f_k(\cdot)}(1^n) = 1] - \Pr_{R \in_R \mathcal{F}_n} [\text{Eve}^{R(\cdot)}(1^n) = 1] \right| \leq \text{negl}(n)$$

Recall: Pseudorandom Functions

HW: Prove Equivalence of Def 1 and 2.

Challenger



$$k \in_R \{0,1\}^n$$

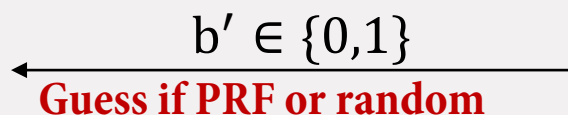
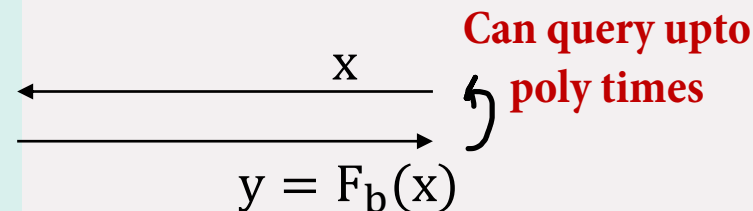
$$R \in_R \{\text{all functions } g: \{0,1\}^n \rightarrow \{0,1\}^n\}$$

Let $F_0 = f_k$

$$F_1 = R$$

Pick $b \in_R \{0,1\}$

$$\mathcal{F} = \{f_k: \{0,1\}^n \rightarrow \{0,1\}^n\}_{k \in \{0,1\}^n}$$



Eve wins if output is 1

$\text{Dist}_{\text{Eve}, \mathcal{F}}$ outputs 1, if $b' = b$ and 0, o.w.

Definition 2

A family of functions $\{f_k: \{0,1\}^n \rightarrow \{0,1\}^n\}_{k \in \{0,1\}^n}$ is a **pseudorandom function** if for every **PPT Eve**, there exists a negligible function **negl** such that, $\forall n$

$$\Pr[\text{Dist}_{\text{Eve}, \mathcal{F}}(n) = 1] \leq 1/2 + \text{negl}(n)$$

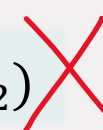
Quiz Time

PRF or not? ($F^{(1)}, F^{(2)}$ are PRFs)

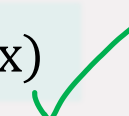
$$F_k(x) = k \oplus x$$



$$F_k(x_1 x_2) = F_k^{(1)}(x_1) F_k^{(2)}(x_2)$$



$$F_{k_1 k_2}(x) = F_{k_1}^{(1)}(x) F_{k_2}^{(2)}(x)$$



HW: Prove this formally.

PRG or not? (F is a PRF)

$$G(s) = F_s(1) F_s(2) \dots F_s(n-1) F_s(n)$$



$$G(s) = F_s(2^0) F_s(2^1) \dots F_s(2^{n-1}) F_s(2^n)$$

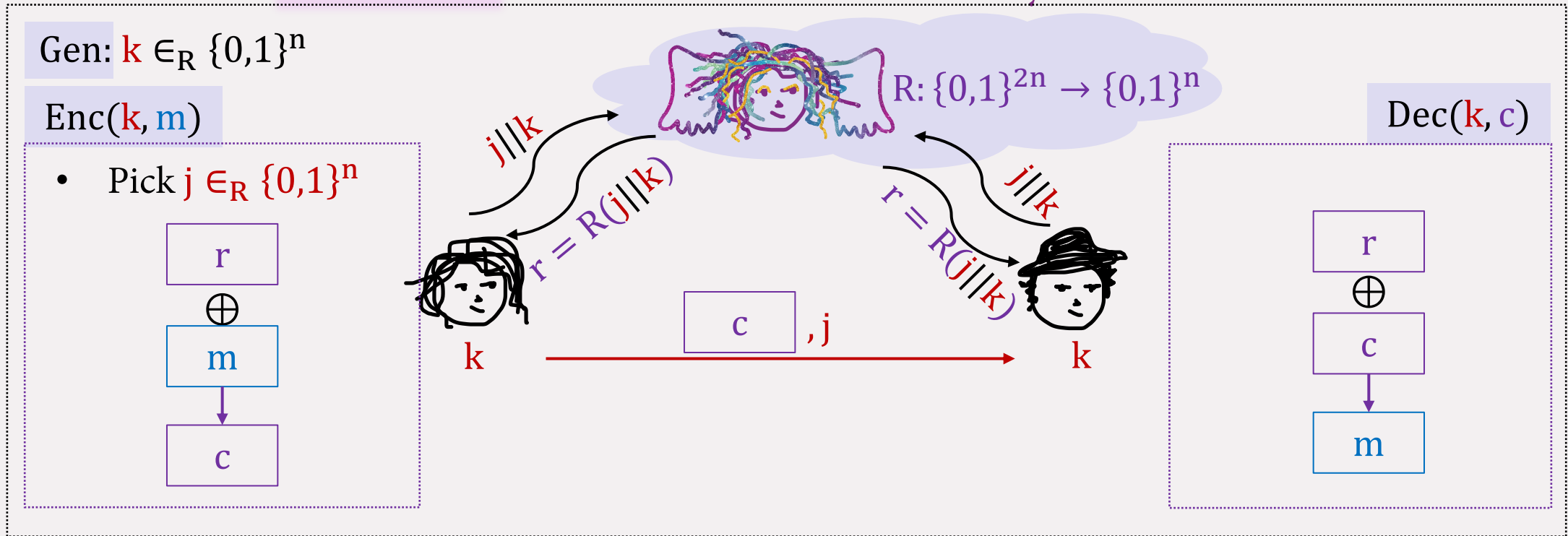


$$G(s) = F_1(s) F_2(s) \dots F_{n-1}(s) F_n(s)$$



Stateless Symmetric Key Encryption

THE POWER: Random Function Oracle in the Sky accessible to all

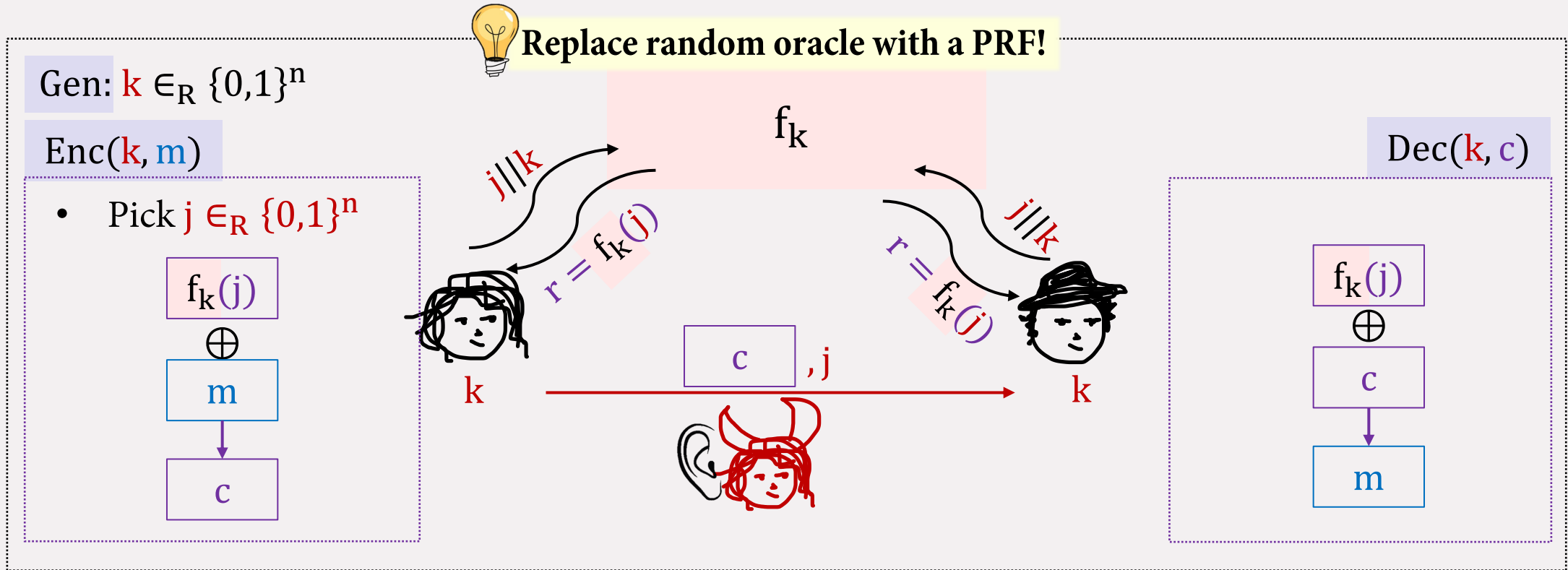


RECALL



Replace random oracle with a PRF!

Stateless Symmetric Key Encryption



Construction 1: Randomized Computational One Time Pad

HW 1: Prove that **Construction 1** is computationally secure for multiple messages against all **PPT** eavesdroppers Eve!

How to build PRFs?

Two Approaches

Theoretical Constructions

Practical Constructions

Rely on well-studied hard problems!

PRGs exist $\Rightarrow$ PRFs exist

[by Goldreich, Goldwasser, and Micali, 1984]

Far from practical!

Start from a design framework
("Appropriately chosen" functions, composed
"appropriately many times" look random)

Build a candidate construction

Do extensive cryptanalysis

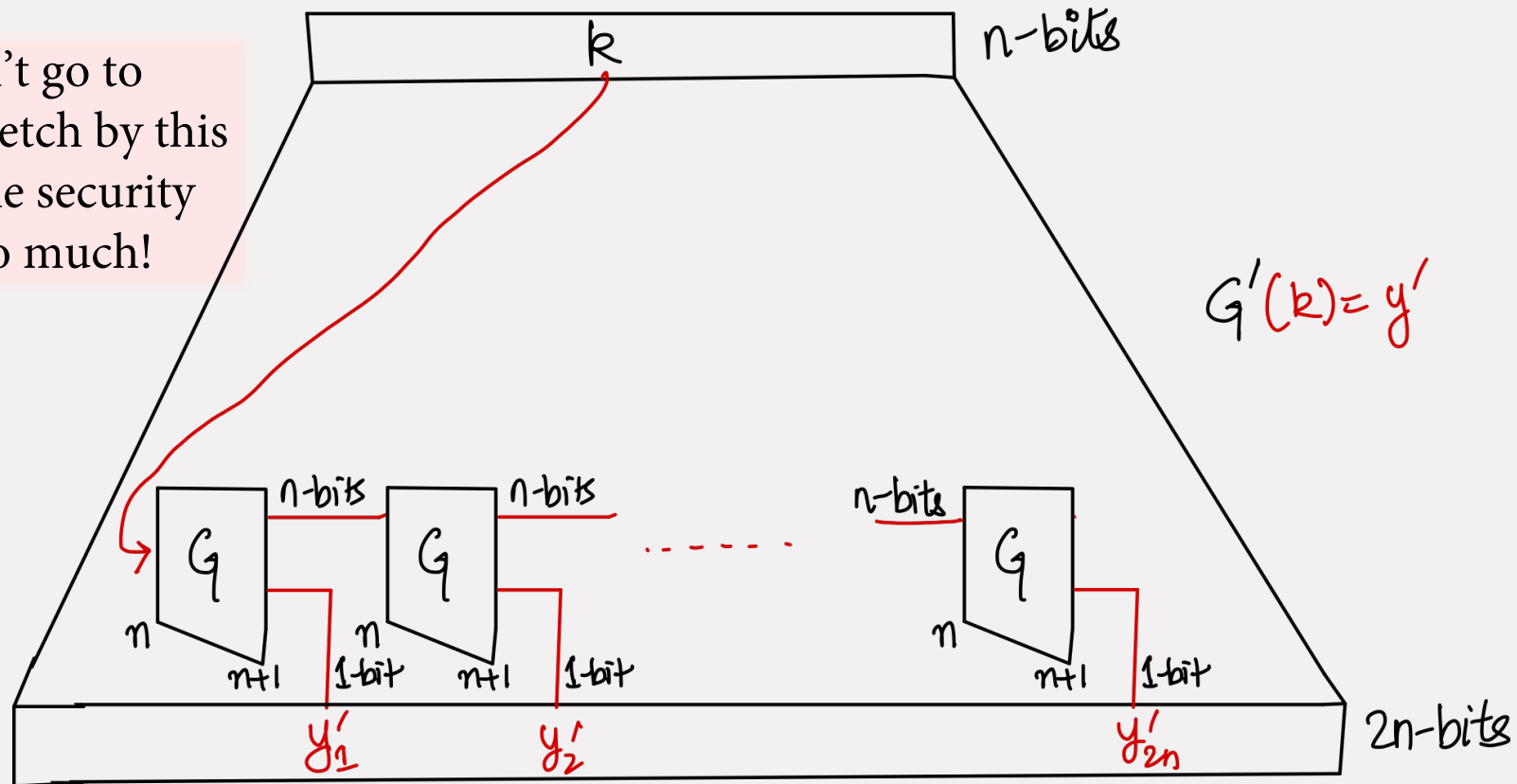
Block Ciphers: AES, DES, etc. [[Boneh and Shoup](#)]

Practical, but not provably secure!

How to build PRFs from PRGs? [GGM]

Recall: The construction of length-extending PRG from last lecture

We couldn't go to exponential stretch by this method, as the security loss was too much!



We need to get to exponential stretch, with efficient accessibility of outputs. How do we do this?

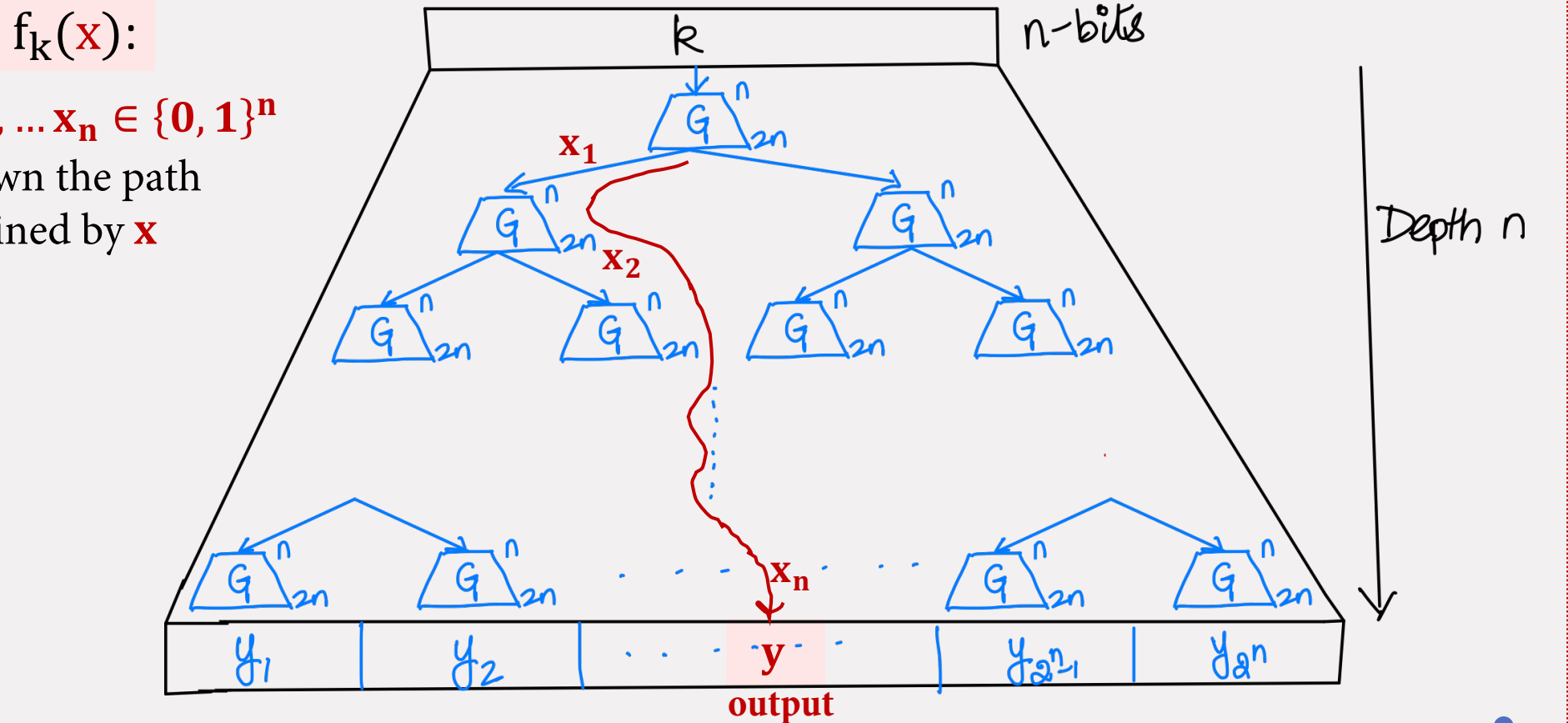
How to build PRFs from PRGs? [GGM]



Use length doubling PRG and use a tree instead of a chain!

$\mathbf{x} = x_1, x_2, \dots, x_n \in \{0, 1\}^n$

Travel down the path
determined by $\mathbf{x}$



Today

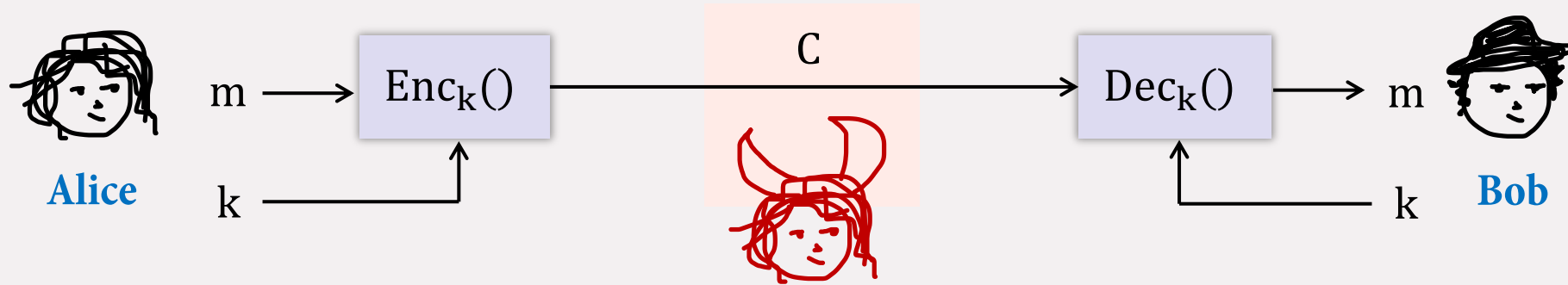


Chosen Plaintext Attacks (CPA)



Modes of Operation

Chosen Plaintext Attack (CPA)



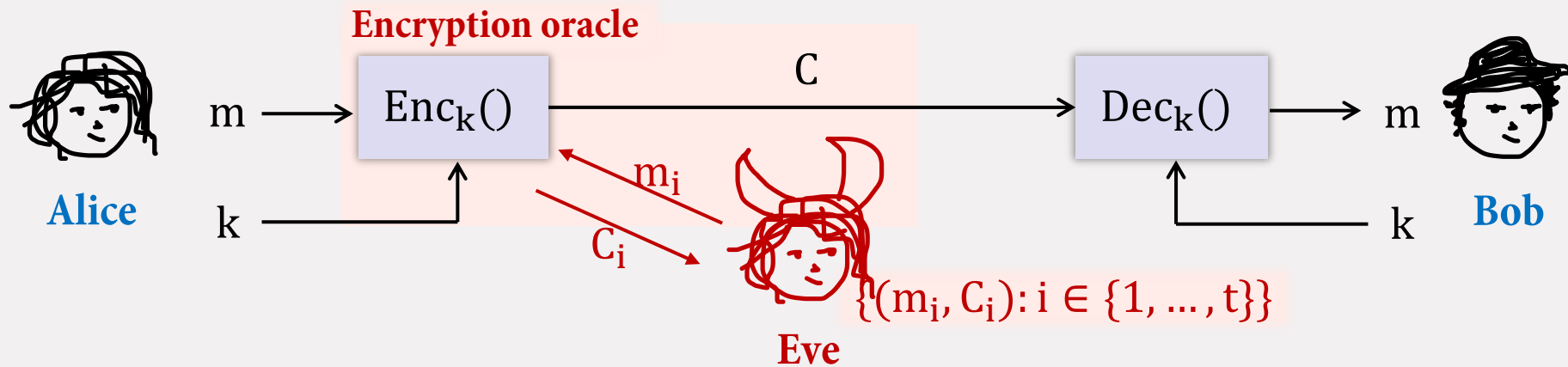
RECALL Eve

Ciphertext Only Attack (COA):

Eavesdropper

PPT

Chosen Plaintext Attack (CPA)

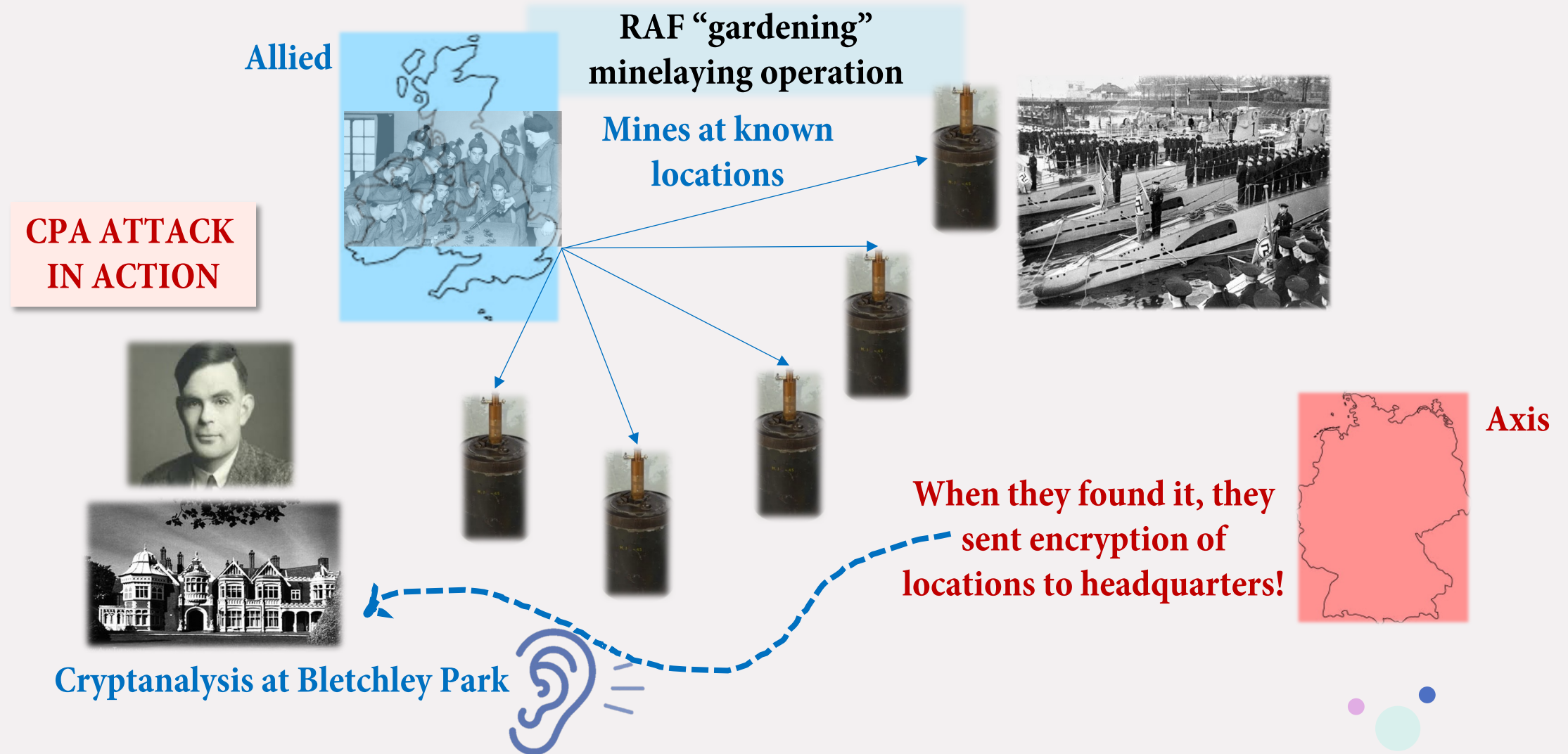


Chosen Plaintext Attack (CPA):

~~Eavesdropper~~ **Eavesdropper with encryption oracle**
PPT

- Eve influences Alice to encrypt messages of her choice (on the same key).
- Eve's goal is then to gain more information about message under a new ciphertext!

CPA shortened WWII by 2-3 years!



Security against CPA



Challenger

$k \leftarrow \text{Gen}(1^n)$

 $b \in_R \{0,1\}$

$\Pi = (\text{Gen}, \text{Enc}, \text{Dec}), \mathcal{M}$

$\text{Enc}_k(\cdot)$ $\xleftarrow{\text{red}} m$ $\xrightarrow{\text{black}} c$

$m_0 \neq m_1 \in \mathcal{M}$
 $|m_0| = |m_1|$

$c_b \leftarrow \text{Enc}_k(m_b)$

$\text{Enc}_k(\cdot)$ $\xleftarrow{\text{red}} m$ $\xrightarrow{\text{black}} c$

$b' \in \{0,1\}$
Guess which one encrypted



PPT

Eve

**Can include
queried messages!**

Eve wins if output is 1

$\text{PrivK}_{\text{Eve}, \Pi}^{\text{cpa}}$ outputs 1, if $b' = b$
and 0, o.w.

Definition 3

Π is **CPA secure** if, for all **PPT** adversary **Eve**, there exists a negligible function **negl** such that, $\forall n$

$$\Pr[\text{PrivK}_{\text{Eve}, \Pi}^{\text{cpa}}(n) = 1] \leq 1/2 + \text{negl}(n)$$

A CPA Secure Scheme

$$\mathcal{K} = \{0,1\}^n, \mathcal{M} = \{0,1\}^n, \mathcal{C} = \{0,1\}^n$$

Building Block

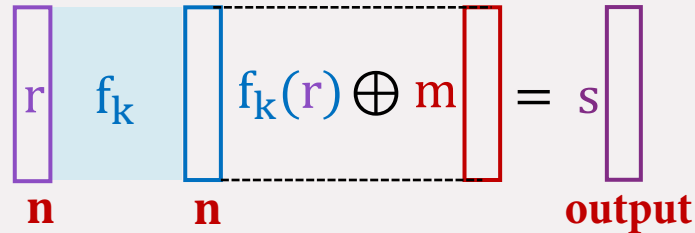
Let $\mathcal{F} = \{f_k: \{0,1\}^n \rightarrow \{0,1\}^n\}_{k \in \{0,1\}^n}$ be a PRF.

Gen(1^n):

- Pick $k \in_R \{0,1\}^n$
- **Output** k

Enc(k, m):

- Pick $r \in_R \{0,1\}^n$



- **Output** $c = (r, s)$

Dec($k, c = (r, s)$):

- **Output** $m = f_k(r) \oplus s$

Theorem 1

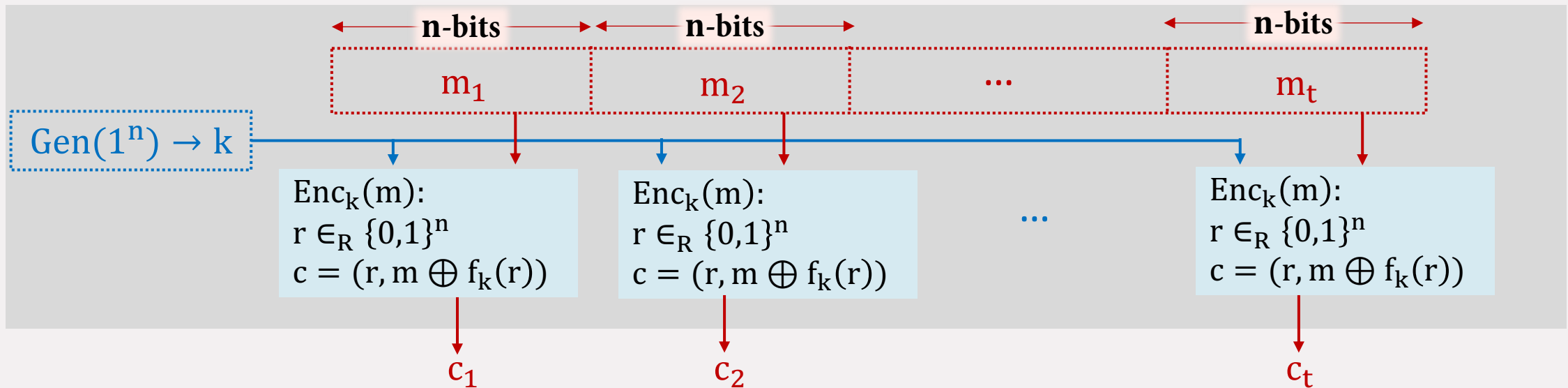
Construction 1

If $\mathcal{F}$ is a pseudorandom function, then **Construction 1** is a CPA secure symmetric key encryption for n -bit messages.

HW 2: Prove this via a security reduction!
[Katz and Lindell second edition, Theorem 3.31]

CPA for Arbitrary Length Messages (Theoretical Construction)

$\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$, $\mathcal{M} = \{0, 1\}^n$ be a CPA-secure encryption for n -bit messages.



Construction 2

HW 3: If Π is CPA secure for n -bit messages then **Construction 2** is CPA secure for tn -bit messages.

[Katz and Lindell second edition, Theorem 3.24]

How good is this?

Message blocks: **t**; Size of each message: **n-bits**

	<u>Theoretical Construction</u>				
Randomness used					
Ciphertext size					
Ciphertext computation parallelizable?					
Randomness reusable?					
Minimal Assumption need					
CPA secure?					

How good is this?

Message blocks: **t**; Size of each message: **n-bits**

	<u>Theoretical Construction</u>				
Randomness used	tn-bits				
Ciphertext size	2tn-bits				
Ciphertext computation parallelizable?	Yes				
Randomness reusable?	No				
Minimal Assumption need	PRF				
CPA secure?	Yes				

How good is this?

Message blocks: **t**; Size of each message: **n-bits**

	<u>Theoretical Construction</u>				Ideal Goal
Randomness used	tn-bits				n-bits
Ciphertext size	2tn-bits				tn + n-bits
Ciphertext computation parallelizable?	Yes				Yes
Randomness reusable?	No				Yes
Minimal Assumption need	PRF				PRF
CPA secure?	Yes				Yes

Today



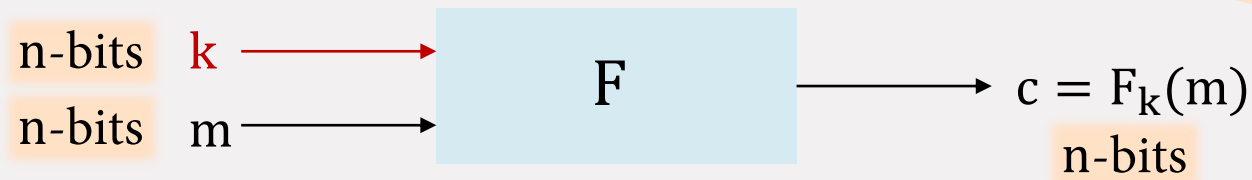
Chosen Plaintext Attacks (CPA)



Modes of Operation

Block-cipher Modes of Operations

- **Given:** A length preserving block cipher F (a PRF (or “**PRP**” or “**SPRP**”)) with block length n



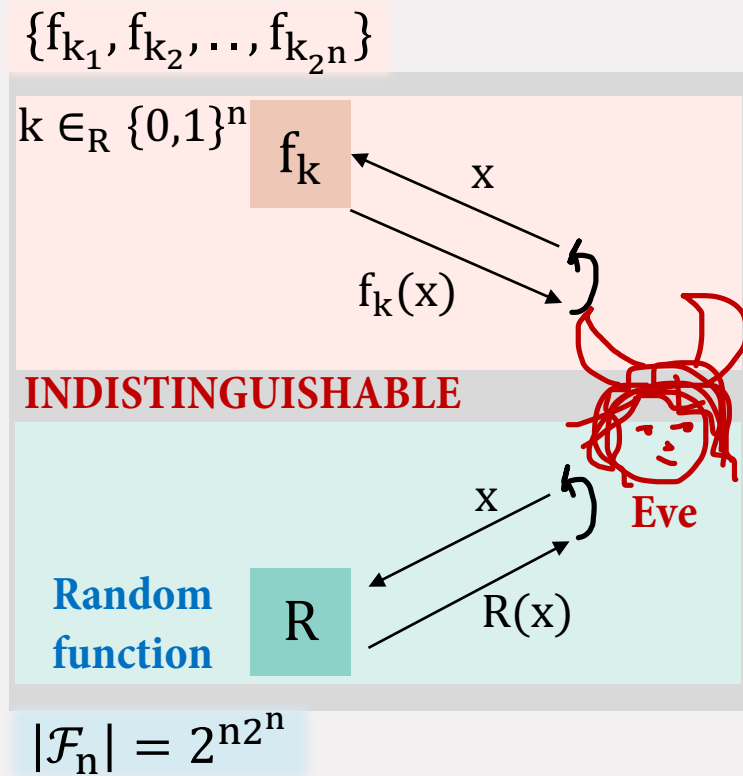
Stronger
variants of PRF

- **Goal:** To encrypt $m = m_1 m_2 \dots m_t$ using F with ciphertext length as small as possible and with randomness as less as possible.

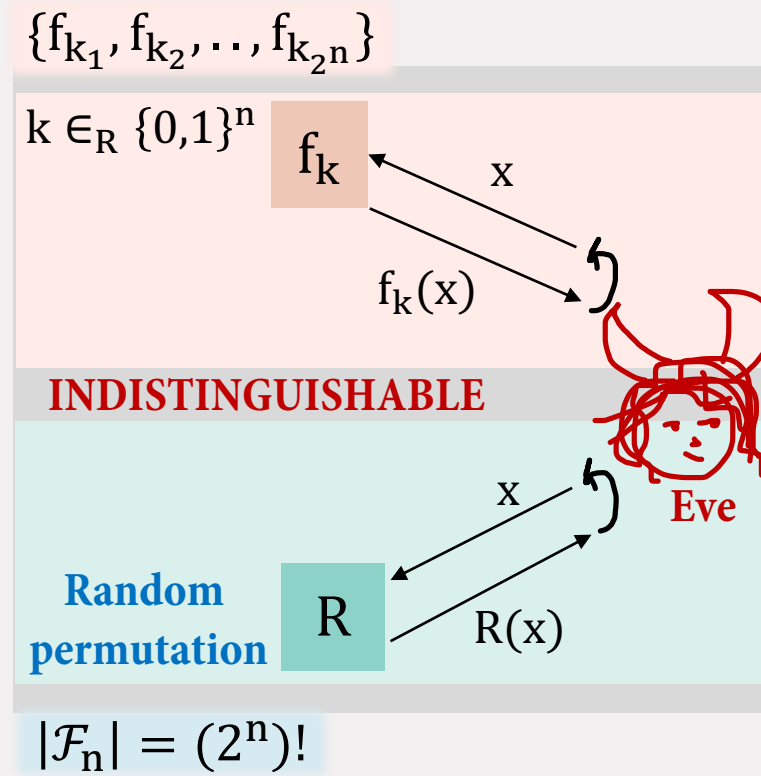


What will you do if messages are not multiple of n -bits long?

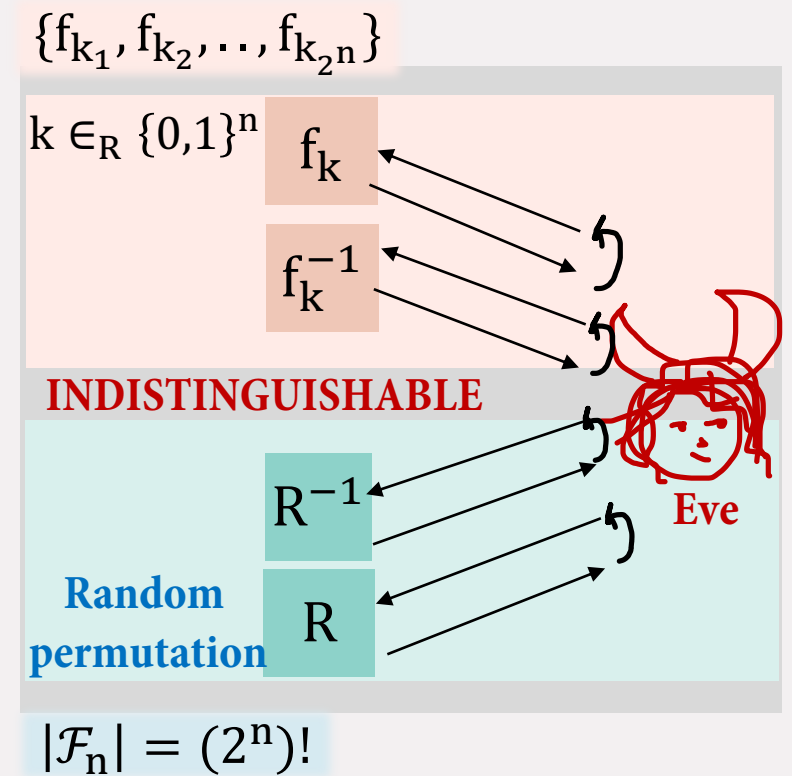
PRP and SPRP



**Pseudorandom Function
PRF**

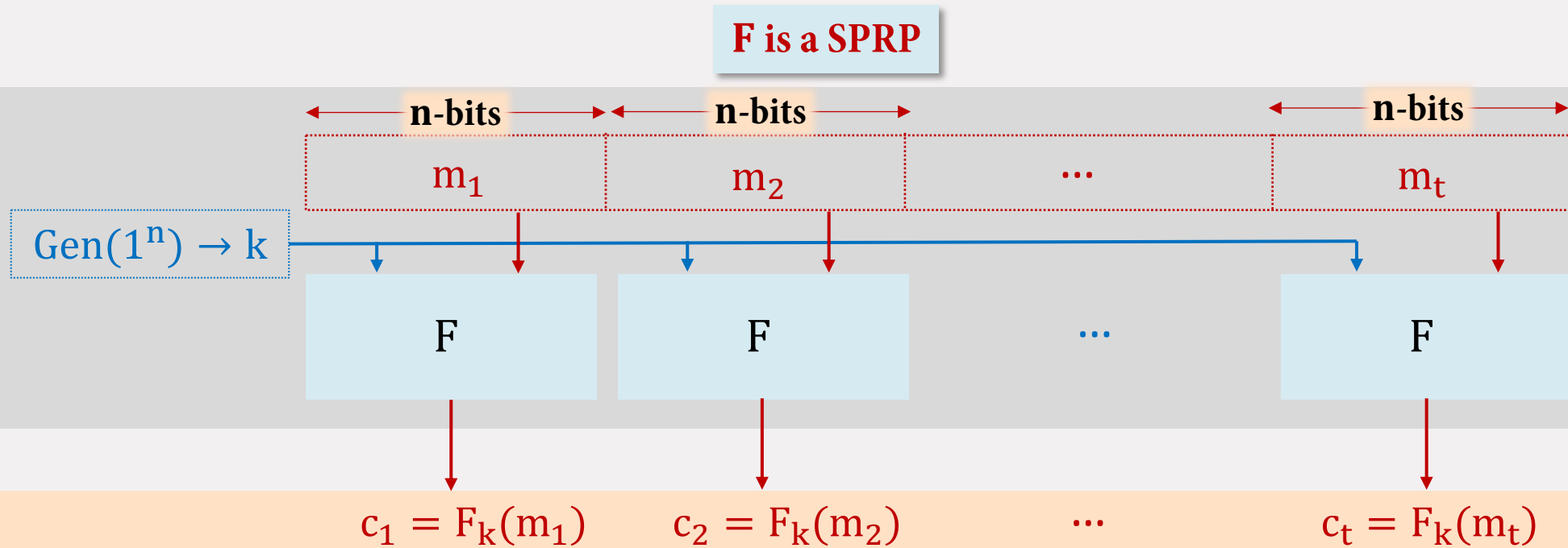


**Pseudorandom Permutation
PRP**



**Strong PRP
SPRP**

Electronic Code Book (ECB) Mode



- No randomness used!
- $|c| = |m|$
- **Decryption:** compute $F_k^{-1}(c_i) = m_i \forall i$
- It assumes that **F is an SPRP**
- Parallelizable? Yes!

CPA SECURITY?



NOT EVEN COA SECURE!

Comparison

Message blocks: **t**; Size of each message: **n-bits**

	<u>Theoretical Construction</u>	<u>ECB</u>			Ideal Goal
Randomness used	tn-bits	0-bits			n-bits
Ciphertext size	2tn-bits	tn-bits			tn + n-bits
Ciphertext computation parallelizable?	Yes	Yes			Yes
Randomness reusable?	No	-			Yes
Minimal Assumption need	PRF	SPRP			PRF
CPA secure?	Yes	No			Yes

Visible Insecurity of ECB

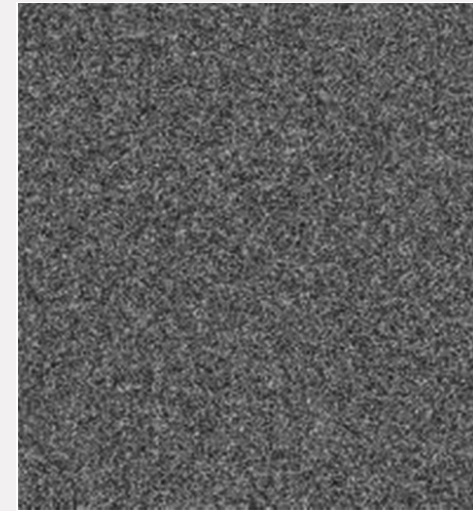
Encrypting an image using ECB, you see the insecurity clearly!



Plain image

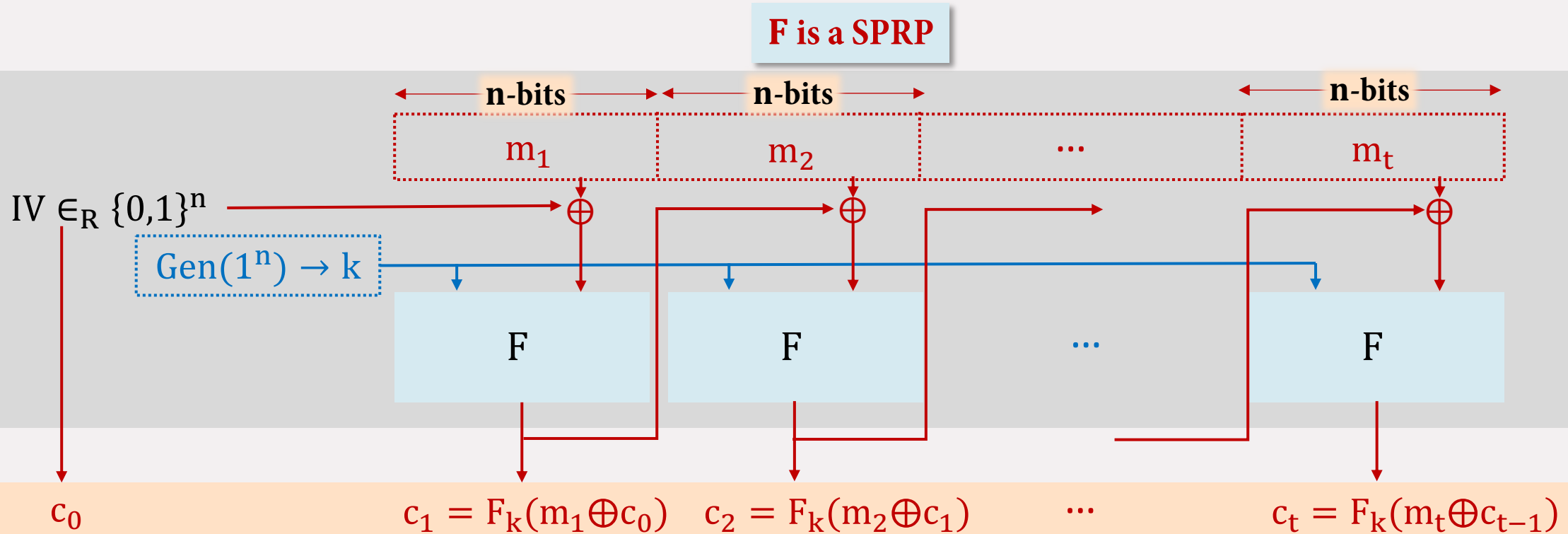


ECB encrypted



Securely encrypted

Cipher Block Chaining (CBC) Mode



- $|c| = |m| + n$
- **Decryption:** $m_i = F_k^{-1}(c_i) \oplus c_{i-1} \forall i$
- It assumes that **F is an SPRP**
- Parallelizable? NO!

CPA SECURITY?

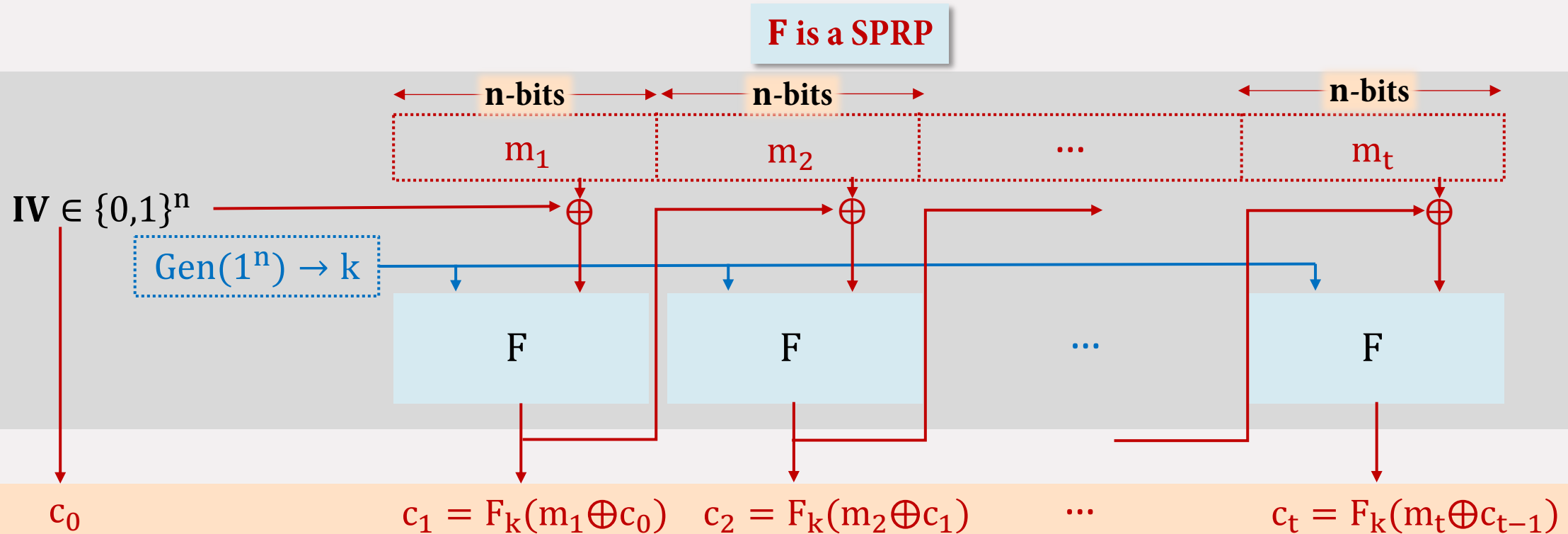
HW 4: Prove it is CPA secure if F is a SPRP!

Comparison

Message blocks: **t**; Size of each message: **n-bits**

	<u>Theoretical Construction</u>	<u>ECB</u>	<u>CBC</u>		Ideal Goal
Randomness used	tn-bits	0-bits	n-bits		n-bits
Ciphertext size	2tn-bits	tn-bits	tn + n-bits		tn + n-bits
Ciphertext computation parallelizable?	Yes	Yes	No		Yes
Randomness reusable?	No	-	-		Yes
Minimal Assumption need	PRF	SPRP	SPRP		PRF
CPA secure?	Yes	No	Yes		Yes

IV misuse in CBC

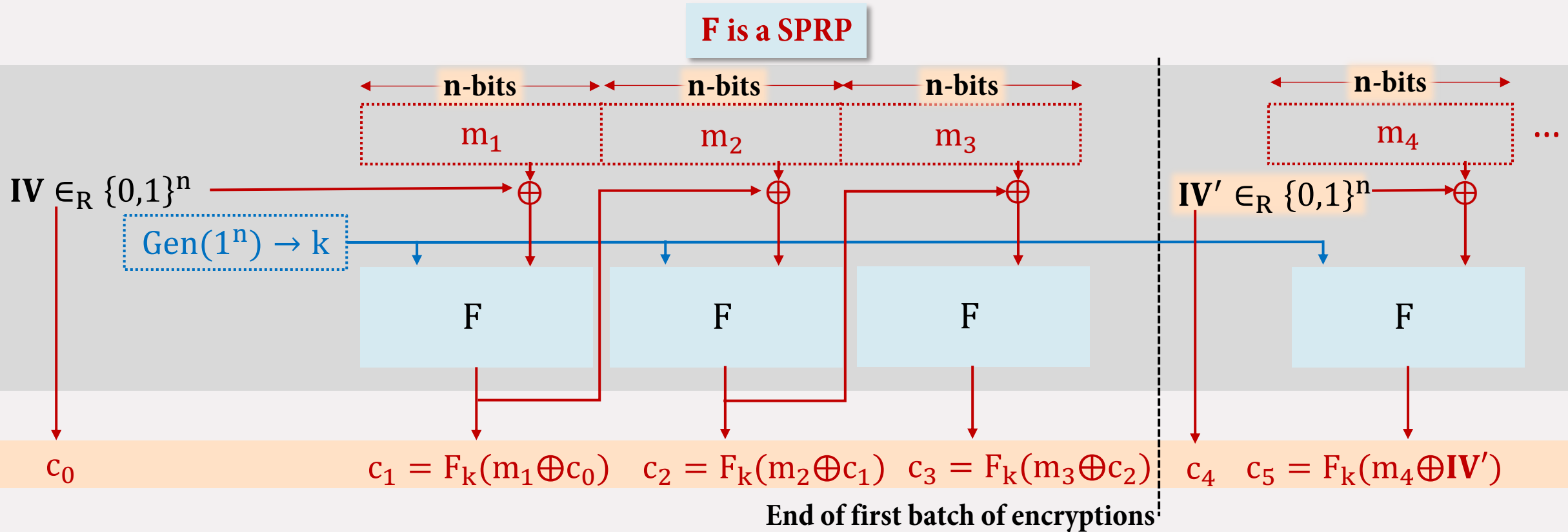


Is choosing a distinct (but not random) **IV** enough (each time a batch of message comes)?

NO!

HW 5: Show that this is insecure!

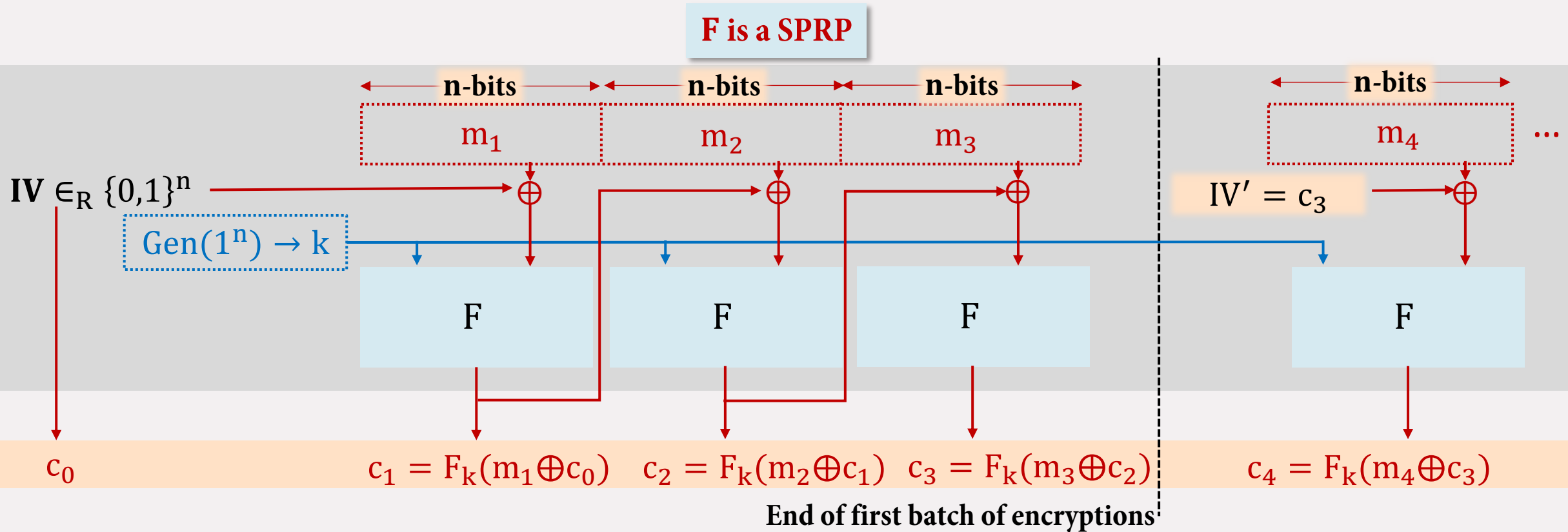
IV misuse in CBC



Stateless CBC Mode

If you had no state information of where you stopped?

IV misuse in CBC



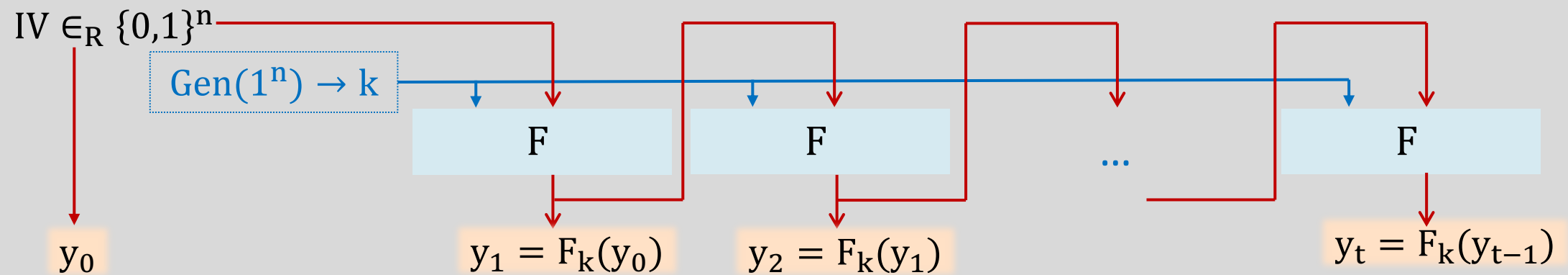
Chained CBC mode –used in SSL 3.0 and TLS 1.0 (stateful variant of CBC)

Can the last ciphertext of previous block be used as IV for the next encryption?
(saves bandwidth and randomness)

**CPA
Attack?**

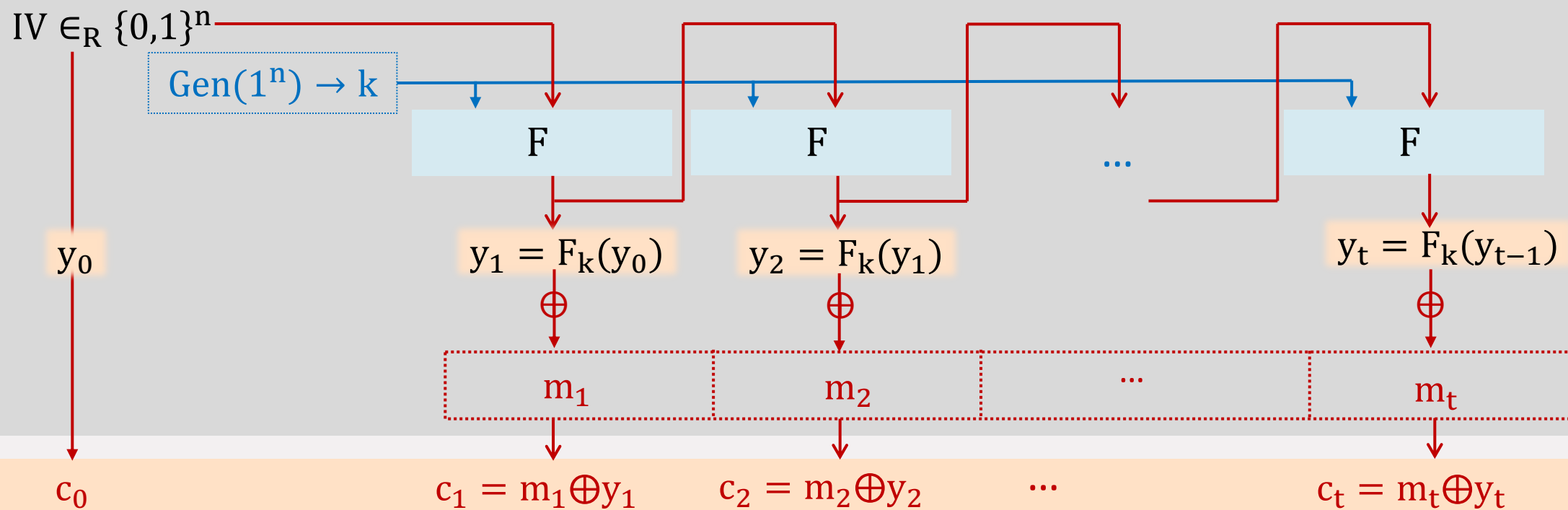
No!

Output Feedback (OFB) Mode



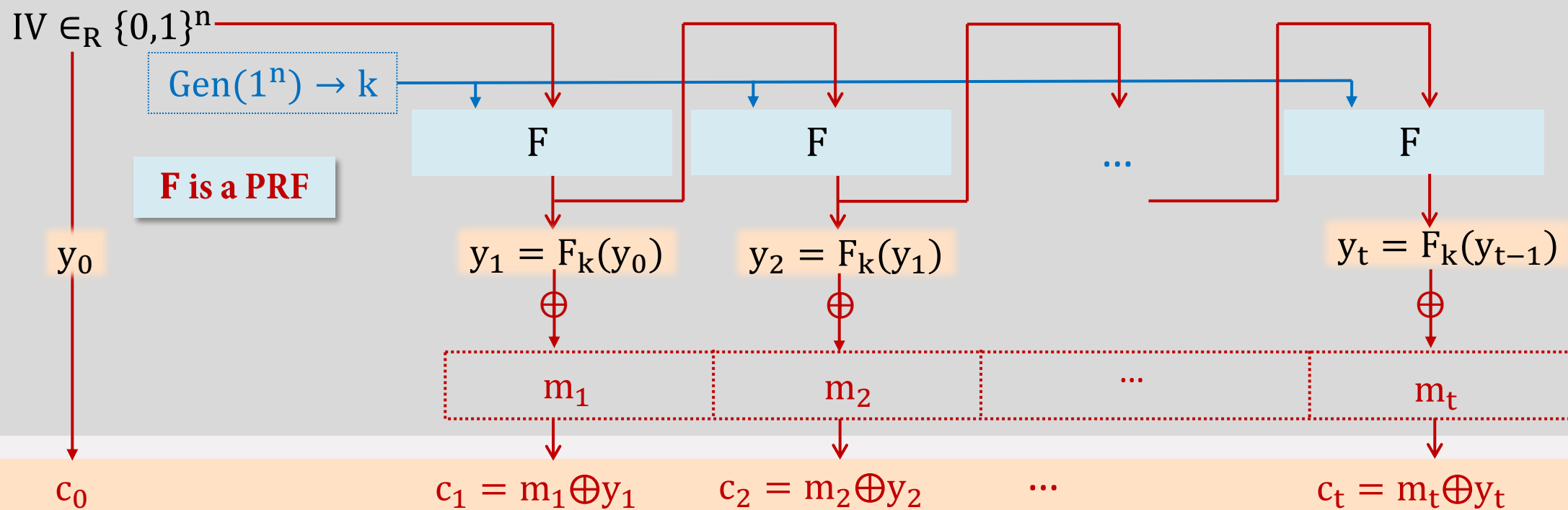
- Generate a pseudorandom stream of pad (independent of message).

Output Feedback (OFB) Mode



- Generate a pseudorandom stream of pads (independent of message).
- Use the pseudorandom stream to mask the message!

Output Feedback (OFB) Mode



- $|c| = |m| + n$
- **Decryption:** $m_i = F_k(y_{i-1}) \oplus c_i \ \forall i$
- Assuming **F** is a PRF is enough!
- Parallelizable? NO, but is precomputable!

CPA SECURITY?

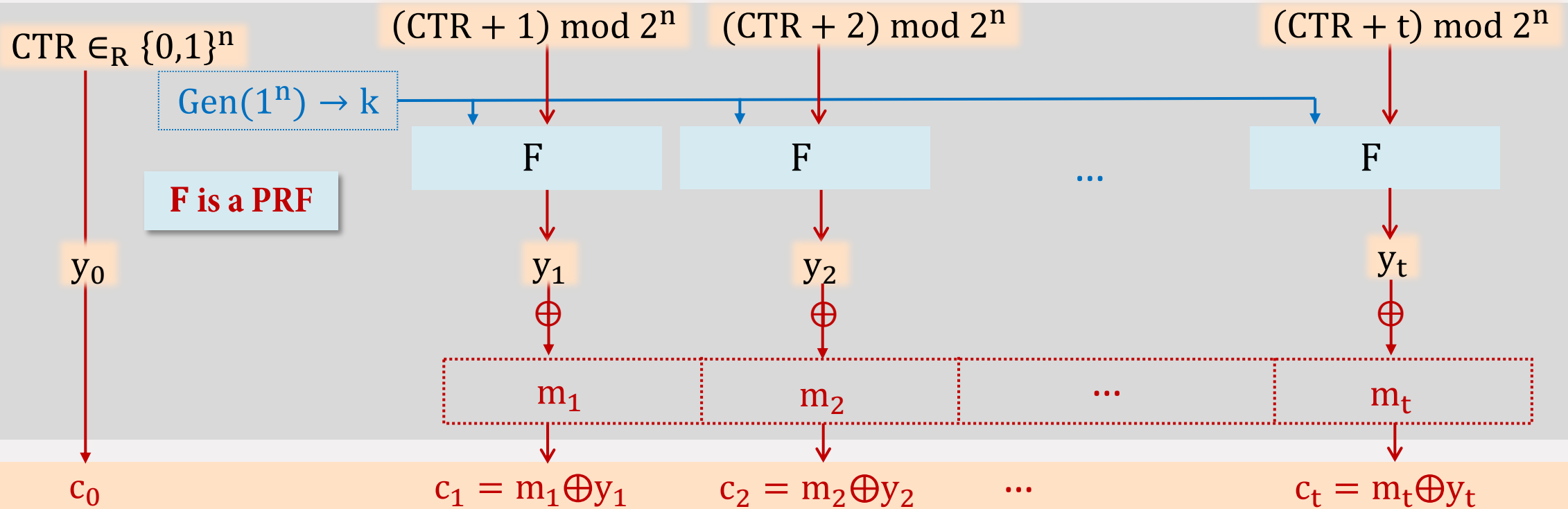
✓ Stateful/chained

Comparison

Message blocks: **t**; Size of each message: **n-bits**

	<u>Theoretical Construction</u>	<u>ECB</u>	<u>CBC</u>	<u>OFB</u>	<u>Ideal Goal</u>
Randomness used	tn-bits	0-bits	n-bits	n-bits	n-bits
Ciphertext size	2tn-bits	tn-bits	tn + n-bits	tn + n-bits	tn + n-bits
Ciphertext computation parallelizable?	Yes	Yes	No	No (but pre-computable)	Yes
Randomness reusable?	No	-	-	Yes	Yes
Minimal Assumption need	PRF	SPRP	SPRP	PRF	PRF
CPA secure?	Yes	No	Yes	Yes	Yes

Counter (CTR) Mode



- It is the same as OFB in structure: pseudorandom stream first, then mask the message.
- But now, parallelizable!
- **PRF enough again!**
- Can decrypt ciphertext block by just invoking F once!

CPA SECURITY?  **Stateful/chained**

Comparison

Message blocks: **t**; Size of each message: **n-bits**

	<u>Theoretical Construction</u>	<u>ECB</u>	<u>CBC</u>	<u>OFB</u>	<u>CTR</u>
Randomness used	tn-bits	0-bits	n-bits	n-bits	n-bits
Ciphertext size	2tn-bits	tn-bits	tn + n-bits	tn + n-bits	tn + n-bits
Ciphertext computation parallelizable?	Yes	Yes	No	No (but pre-computable)	Yes
Randomness reusable?	No	-	-	Yes	Yes
Minimal Assumption need	PRF	SPRP	SPRP	PRF	PRF
CPA secure?	Yes	No	Yes	Yes	Yes

Practical Issues

1. Block Length and Concrete Security

- If F supports block length of n -bits, then $IV \in_R \{0,1\}^n$
- After $2^{n/2}$ encryptions, IV will repeat with constant probability! 
- If n too short, then impractical security!

Example: DES with $n = 64 \Rightarrow$ repetition after $2^{32} \approx 4,300,000,000$ encryptions!
(approx 34 GB of plaintexts —not too large)

2. IV misuse

- IV repeated then:
In **CTR** and **OFB**, same pseudorandom string generated –security lost!
In **CBC mode**, effect is not that serious –Why?

↓
To avoid this, either use **CBC** or stateful **OFB/CTR**

Practical Issues

1. Message Transmission Errors (non-adversarial)

- Dropped packets, changed bits, etc.
- Solved by some standard techniques: re-transmission or error correction

2. Adversarial Tampering

- What if Eve now is allowed to modify ciphertexts?
- Issue of message integrity/authentication!

NEXT WEEK

LAB EXERCISE 2: out on 29th Aug



THANK YOU